

Thomas Fieguth

B.A.Sc Systems Design Engineering, University of Waterloo – Dean's Honour Roll
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Skills

- Languages: Python, C++, Kotlin, ROS2, Arduino
- Frameworks & Tools: ROS2, Qt (GUI & 3D visualization), SolidWorks, CAD, CNC programming
- Fabrication: Woodworking, welding, soldering, machining, CNC milling, programmable lathes
- Languages: English (fluent), German (fluent), French (immersion)

Experience

Werkr Tractors – Software Engineer May 2026 – Present

- Full-time software engineering role following co-op conversion

Werkr Tractors – Software Engineer Co-op May – Aug 2025

- Tested and validated autonomous tractor system, improving navigation accuracy and row-following performance using camera-based plant detection
- Refined onboard Qt user interface for tractor control

Werkr Tractors – Robotics / Software Engineer Co-op Jan – Apr 2024

- Developed autonomous robot navigation in C++ and ROS2 by fusing multiple sensor inputs for path following
- Built onboard control and visualization tools using Qt, including a GUI and 3D object viewer

GEI Consultants, Inc. – Project Designer Co-op May – Aug 2023

- Analyzed pump efficiency and friction losses in piping systems
- Developed a capacity-based cost estimation model for water and wastewater treatment plants
- Conducted site inspections and authored formal client reports

Accelerated Systems Inc. – Mechanical / Software Engineer Co-op Sep – Dec 2022

- Designed and developed a Kotlin mobile app to wirelessly control an autonomous lawnmower and visualize real-time sensor, GPS, and camera data
- Implemented ROS2-based C++ control systems for robot motion
- Machined and assembled motor casings for higher-efficiency golf cart motors

Accelerated Systems Inc. – Machinist & Technical Support Co-op Jan – Apr 2022

- Designed motor testing stand and dynamometer for bike motors using SolidWorks
- Programmed CNC machines for custom part fabrication and rapid prototyping

Projects

- Depth sensing real time spatial feedback for learning woodcarving (capstone)
- Topographical map of farm with different wood types
- CNC-routed wooden globe
- Go-kart electric conversion
- Stormbreaker axe
- Designing Custom Lego sets

Education

B.A.Sc Systems Design Engineering – University of Waterloo 2021 – 2026

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